CHRISTIAN OMAR PAYÁN TORRÓNTEGUI

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SUMMARY

Robotics & Digital Systems Engineering student focused on product design, advanced manufacturing, and process improvement. Proven experience in 3D modeling, additive manufacturing, and embedded systems. Seeking to leverage mechanical design to bridge creative design with technical feasibility and optimize industrial processes.

EDUCATION

B.S. Robotics and Digital Systems Graduating Expected December 2026
Tecnológico de Monterrey
Zapopan, Jalisco
Relevant coursework: Industrial Mounts, Process Optimization, Agile methodologies

TECHNICAL SKILLS

Programming & Languages: Python, C++, Spanish (Native), English (B2)
Robotics & Simulation: ROS 2, Gazebo, URDF
Design & Electronics: SOLIDWORKS, Fusion 360, Eagle
Office & Productivity: Microsoft Office (Excel, Word, PowerPoint, Outlook — 100-hour certification)
Tools & Methodologies: Git, Agile, Scrum, Advanced FDM 3D Printing, Linux (Ubuntu), Docker

PROFESSIONAL EXPERIENCE

Industrial Mounts & Robotic Support Interfaces, Zapopan, Jalisco: Mechanical & Product Designer Feb-Jun 2026

- Engineered mechanical interfaces for VR controllers, developing a 3-DOF system that reduced fabrication costs by 95% vs commercial alternatives
- Designed and prototyped a custom mini PC enclosure for robust deployment in industrial environments, optimizing material choices for thermal and mechanical properties

ACADEMIC & TECHNICAL PROJECTS

Open-Source Gastrointestinal Medical Phantom Ago-Dic 2025 – Feb-Jun 2026
Agile, Advanced FDM

- Engineered phantom via advanced FDM 3D printing (TPU/PETG), reducing prototyping costs by 80% vs commercial alternatives
- Managed development using Agile methodologies to iterate rapidly on anatomical accuracy and material constraints

Port Automation & Drone Sim Ago-Dic 2025
ROS 2, Gazebo, C++, Python

- Engineered URDF models and C++/Python ROS 2 nodes for kinematic control and reliable flight telemetry
- Designed & 3D-printed custom hardware: drone mirror/lens mounts, and an AMR bracket (LiDAR, ArUco, Jetson)

DigiMix Ago-Dic 2024
Embedded Systems, Scrum

- Designed custom PCB and 3D-printed casing; engineered hardware using dual-core STM32 and ESP32 (WebSockets)
- Applied Scrum to manage lifecycle, integrating hardware and software architectures with zero logic faults